ARQUIVISION AI

Generative-AI rendering studio for architects and interior designers.

From a client's photo to a photoreal redesign - in seconds, in their words.

LIVE - bednrixvision.ai

Final Project - Generative AI - MCSBT, IE University

Vintimilla - Ortega - Seben

THE PROBLEM

Rendering is the bottleneck in every design pitch

A studio cannot show a client what a change looks like while the conversation is still alive. One photoreal render goes through a 3D artist - and the momentum dies between meetings.

EUR 150-600

cost of a single room render

2-5 days

turnaround through a 3D artist

1 meeting

the window you have to close the sale

THE SOLUTION

Collapse days into seconds - in the client's own words

The designer uploads the client's real photo, types the change in plain Spanish or English, and gets a photoreal variation instantly - iterating live until the client says "that one."

Without ArquiVision

- Brief the 3D artist, wait days
- EUR 150-600 per room, per style
- Client decides between meetings - or not at all

With ArquiVision

- Upload the photo, type the change
- Under 30 seconds, cents per image
- Decision closed in the first meeting

Replace a EUR 150-600, multi-day task with a sub-30-second, sub-$0.30 generation.

WHO IT'S FOR

Built for the people who sell design



Architecture & interior studios

The core user: redesign a client's space live, in any style, during the meeting.



Real estate & virtual staging

Furnish and restyle empty or dated properties to sell or rent them faster.



Developers & reformers

Show buyers a finished look from a bare floor plan before a single wall is built.

THE PRODUCT

Four workspaces, one design session

Photo Enhancer

Upload a room photo, instruct edits in natural language. Style presets, masking, reference-image transfer, A/B lab.

Plan to Render

Turn a floor plan (PNG or DXF) into a furnished photoreal interior, or edit the plan with vector operations.

Video Studio

Turn any render into a cinematic architectural fly-through with geometry and lighting preserved (Veo 3.1).

Admin Dashboard

Live usage, latency and cost telemetry per model and per day - the basis for pricing the product.

WHY IT'S NOT JUST AN API WRAPPER

The model is swappable. The intelligence around it is the product.

1

Classify intent

Spanish architectural vocabulary - cornisa, moldura, cuarteron, zocalo - mapped to a precise edit type.

2

Analyze scene

Read the exact camera, lighting and layout of the original, and cache it for the whole session.

3

Generate

Tell the model what to change and what to keep, then output a photoreal variation.



Spanish vocabulary engine



Quality tiers + fallback



Cumulative edit lineage



Per-user cost guardrails

ARCHITECTURE

A secure proxy between the studio and the models

Frontend - React + Vite

Photo Enhancer
Plan to Render
Video Studio
Admin Dashboard
Supabase auth (JWT)

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Backend - Flask proxy

Keys stay server-side
JWT validated per call
Tiered model routing
Daily spend cap
Usage logging

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Model layer

OpenAI gpt-image-2
Gemini 2.5 / 3 image
Gemini 2.5 flash (vision)
Veo 3.1 (video)
Fallback chain

Supabase - Postgres + Storage + Row-Level Security

usage_logs (model, latency, cost) - creative_datasets (learning signal) - conversations & render gallery - RLS-gated inserts run as the user

TECHNICAL DEPTH

Quality tiers map to model, resolution and batch

Tier	Provider - Model	Resolution	Batch	Use
Draft	Gemini 2.5 Flash Image	default	1	Sub-3s iteration
Standard	OpenAI gpt-image-2	1K	2	Pick-from-variations
HD	OpenAI gpt-image-2	2K	2	Client-ready render
Ultra	OpenAI gpt-image-2	4K	1	Hero deliverable



Silent fallback: OpenAI to Gemini Pro to Flash on any 429/503.



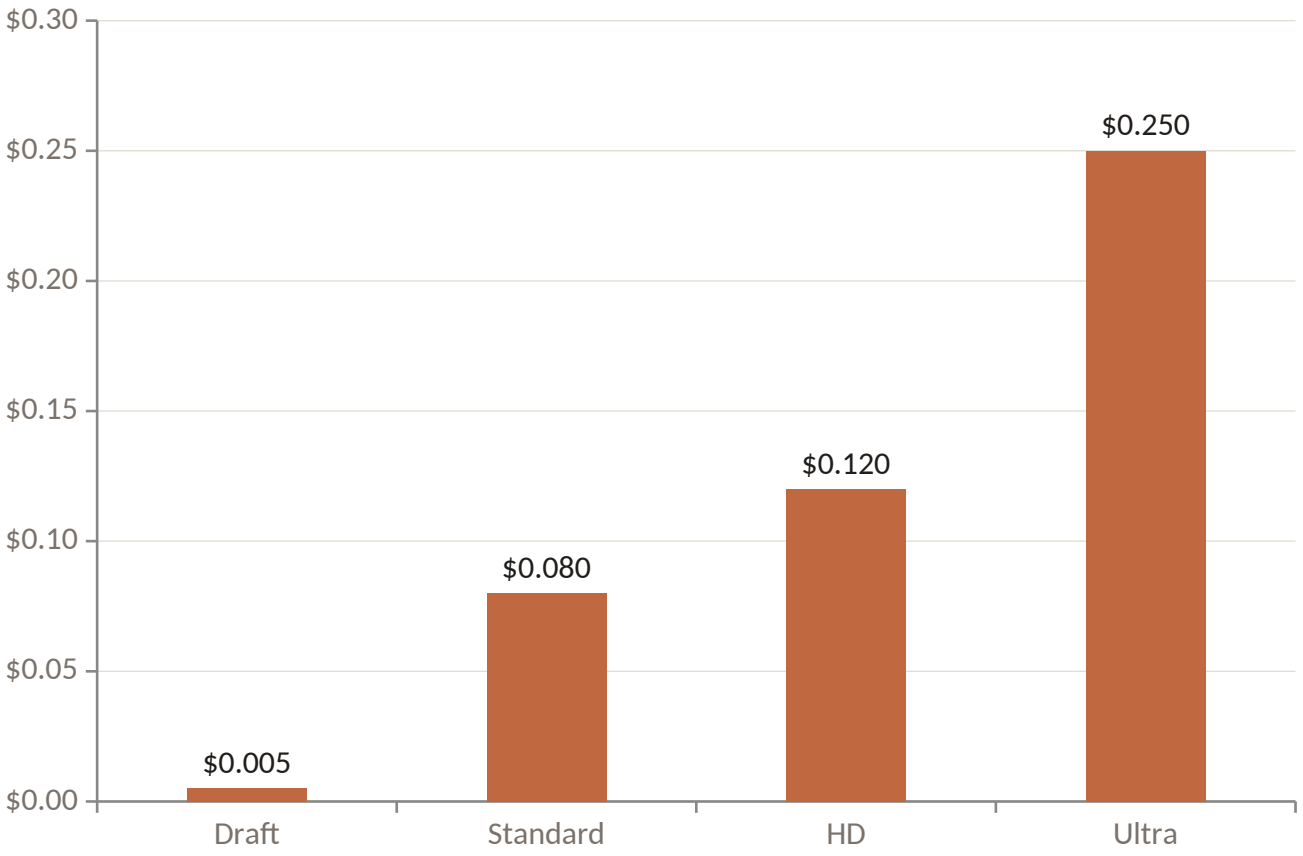
gpt-image-2 reasoning loop + native 2-image batches with object continuity.



Every call logged with model, latency and estimated cost.

UNIT ECONOMICS & GUARDRAILS

Cents per image, with a hard daily cap



vs EUR 150-600

A studio render through a 3D artist. ArquiVision delivers the same client-facing visual for a few cents.

Daily per-user spend cap

Beyond the cap, generation drops to the free Draft tier and the UI warns the user. One power user can't run up the bill.

WHAT'S BUILT

A working MVP, not a prototype

- End-to-end: upload to instruct to photoreal render in the browser
- Image to video fly-through (Veo 3.1)
- Natural-language editing with Spanish architectural vocabulary
- Supabase auth, persistent gallery & conversation history
- Masking, reference-image style transfer & multi-view input
- A/B lab: gpt-image-2 vs Gemini, side by side
- Plan to Render with geometry conditioning (PNG / DXF)
- Cost telemetry + daily spend cap in the Admin Dashboard

ROADMAP

From working MVP to studio-grade product

Shipped	Next	Then	Later
gpt-image-2 swap, edit lineage, multi-view, Plan to Render, A/B lab	Flux + ControlNet for locked plan geometry (no wall drift)	Real upscaler for Enhance HD + a video quality tier	Multi-page client PDF deliverables & smart style suggestions

Detailed engineering plan lives in the repo: .agents/arquivision_improvement_plan.md

ArquiVision AI

Design decisions, closed in the first meeting.

Live demo: bednrixvision.ai

github.com/bendervintimilla/bender-archivisionAI

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Thank you - questions?